

University at Buffalo
Department of Computer Science & Engineering
338 Davis Hall – (716) 645-3180

Syllabus

Please read this sheet carefully, and save it for future reference.

The syllabus is subject to change based on the needs of the course and will be communicated with you as appropriate.

Instructors

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Teaching Assistants

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Office Hours

Most up to date office hours will be posted on piazza in a pinned post. Office hours will be a mix of zoom and in person to offer the most flexibility and to account for student preferences. We will adjust office hours based on student needs and hold additional office hours to help everyone prepare for final project submissions, midterms, and final exams.

Course Information

CSE 305 Introduction to Programming Languages – 4 credits

Course Website: piazza.com/buffalo/spring2025/cse305/home

Prerequisites

You must have passed CSE250 and be an admitted CSE student, or have an equivalent background and receive permission from the instructor.

Course Description

Examines concepts and constructs found in programming languages. Topics include binding time, strong typing, control and data abstraction, higher-order functions, and polymorphism. The major programming paradigms (procedural, object-oriented, and functional) will be studied and compared. The course will also provide an introduction to syntax and semantics, and compilation vs. interpretation.

A day-to-day listing of lecture topics and required readings is available on the course web site (Top Hat) and Piazza.

Objectives and Outcomes

The course covers topics from the following knowledge units from ACM's CC2001 curriculum document: **PL1** Overview of programming languages, **PL2** Virtual machines, **PL3** Introduction to language translation, **PL4** Declarations and types, **PL5**

Abstraction mechanisms, **PL6** Object-oriented programming, **PL7** Functional programming, **PL9** Type systems, **PL10** Programming language semantics, and **PL11** Programming language design.

This course is required of all computer science BS students and has a significant relationship with the following program objectives:

CS-1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.

CS-6: Apply computer science theory and software development fundamentals to produce computing-based solutions.

Learning Outcomes and Program Objectives Mapping

Upon successful completion of this course a student will...	CS-1	CS-6	Assessment Instruments
Be able to reason about the mathematical underpinnings of programming languages.	✓	✓	Participation
Be able to implement a functional language with support for first class functions, recursion, and basic data types.	✓	✓	Project
Be able to exploit knowledge of the essential structure of programming languages to learn a new programming language.		✓	HW, Project
Be able to implement solutions to non-trivial programming problems in a functional programming language.		✓	HW, Project
Demonstrate understanding of different language features and how they affect the semantics of the languages (e.g. name resolution, calling conventions, repetition, sequencing).	✓		Participation
be able to characterize the trade-offs in security /execution speed / programmer flexibility in compiler-implemented, runtime-implemented and programmer-implemented features, such as type checking, bounds checking, and memory management.	✓		Participation

Textbook and Materials

There is one textbook for this course. It is suggested you have it, but it is **not required** (i.e. the book is optional). If you learn by reading, or find written material useful for review, it is suggested you purchase the textbook.

- Robert W. Sebesta. 2012. *Concepts of Programming Languages* (10th ed or 11th ed)*, Addison Wesley. (ISBN: 978-0-13-139531-2).

*11th edition is the preferred version of the book, but the 10th will work as well

Additional reading material may be assigned during the course, and will be announced in lecture and posted on Piazza.

Computing Resources

This course will leverage Top Hat, Piazza, UBLearn (BrightSpace), and AutoLab. A subscription to Top Hat is required for the course (link to Top Hat and join code are available on UBLearn).

You will be provided with a CSE undergraduate computing account. *It is your responsibility to ensure that any programs you write for this course compile using the compilers/interpreters installed on the department's machines.* Information about the CSE computing environment can be found at

<https://wiki.cse.buffalo.edu/services/>

Any email communication that you send regarding this course must be sent from your UB email account.

Course Organization

The course has both a lecture component and a recitation component. Each component plays a role in helping you achieve the objectives of the course. **If you do not participate fully in both you should not expect to do well in the course.** Outside of the scheduled course times, both office hours and your own study times are critical components of the course.

Lectures

The conceptual and theoretical course content will be delivered primarily in the lectures, complemented by readings from the textbook. You must review readings prior to attending a lecture, and you are expected to review the readings again, along with any notes you took, after the lecture. Lectures for this course will be given live, and pre-recorded videos will also be made available.

Some of the topics will be difficult. It is therefore absolutely essential that you ask questions whenever something is said which you do not understand. We are here to help you work through question you have, so please utilize office hours, ask questions during lectures and recitations, and post to Piazza.

Recitations

The recitations are an integral part of this course as they provide a smaller group environment to ask questions and work through problems. *Attendance in all recitations is critical to your success in the course. Attendance will be taken in recitations via TopHat.*

Top Hat

Top Hat is required for this course. Once you sign up for Top Hat, you will notice that we have arranged the course in pages, one page for each week. This page is effectively a guide for that week's course content. You will find videos, quiz questions, links to resources, as well as non-required additional material that you may find useful. This weekly page is your one stop shop for 305 for that week. We have broken up videos into small digestible chunks so you do not need to sit and watch hour long videos one at a time. Instead, videos will be roughly 5-15 minutes long depending on content. A few of them are longer, but most are short. After each video or sometimes after a pair of videos you will find top hat questions you can take as a "self-quiz". **These questions will count toward your participation grade.** You will also find short recitation videos, either working through problems or as "live coding" examples. This means instead of having two to three lectures and 1 recitation in monolithic videos, instead we will have many shorter videos and recitation will be interleaved with traditional lecture based around concepts. The videos provide the same material covered in the live lecture and recitations, just in a different format.

Time outside of class

Office hours

Office hours offer you the opportunity to ask more individual questions about the course material than can typically be addressed in lecture. Both the instructor and the teaching assistants have scheduled office hours. Office hours are held on a first-come first-served drop-in basis. No appointment is necessary to attend office hours. Be aware that office hours become increasingly busy the closer it is to a deadline or exam. Plan your use of office hours accordingly.

Individual appointments may be arranged, if needed, as schedules allow.

Study time

In this course, as in any course, you are expected to put in additional time beyond the scheduled class times. Professors generally expect that for each credit hour a class carries a typical student will put in 2 – 3 hours of time **each week** outside of class. **Since this is a 4 credit course that translates into 8 – 12 hours of time outside of lecture and recitation times, each week.** During this time you should review your lecture notes, attend office hours as needed, get hands-on practice applying the concepts and theoretical constructs discussed in class, prepare to code up your homework, and work on the course project.

As a rough guide, you should expect to spend *at least* the following time working on this course, *each week*:
lectures (3 hours) – recitation (1 hour) – programming assignments (4 hours) – individual study (3 hours)

Grading Policy

The following indicates the grade breakdown, which will be used in assigning grades in the course. We reserve the right to make adjustments if we deem it to be necessary. Any changes will be communicated to the class in writing via email to each student's UB email account.

1. Programming component (50% of final course grade)***Project Component***

There will be a course project, which is split into three parts. The purpose of this is to reinforce and deepen your understanding of the broader concepts discussed in class through application of those concepts to concrete problems. The project serves as a capstone for all concepts in the course. You will be implementing a programming language.

The project parts will all be due on the same day. However, suggested due dates will be provided so you can pace your completion of the project. Each project part will be graded separately and will be due no later than 11:59 PM on the day the project is due, unless adjusted on the course website. All project parts are submitted on-line through AutoLab.

It is your responsibility to ensure that any programs you write for this course compile using the compilers installed on AutoLab. **Submissions that do not compile will be given a zero grade.**

The course assignments and project are to be completed by yourself. You may, however, discuss high level ideas on the course Piazza page. **Source code may not be shared or posted.**

Late policy for project submissions

Late homework submissions are not accepted, however all due dates include extra time. This extra time is meant to provide you with flexibility on how to manage your time. We will use AutoLab for grading and you may submit as many times as you want prior to the due date. This also allows you to immediately know your grade. We encourage you to submit early and frequently.

Homework component

There will be 3 programming assignments in addition to the course project. The purpose of these is to reinforce and deepen your understanding of the broader concepts discussed in class through application of those concepts to concrete problems. The programming assignments are designed to give you hands-on experience analyzing problems, developing solutions to them, and implementing these solutions in the languages we are studying in depth. These assignments also serve to give you feedback on your understanding of the material.

Assignments will be due no later than 11:59 PM on the day they are due, unless adjusted on the course website. Assignments are submitted on-line through AutoLab.

It is your responsibility to ensure that any programs you write for this course compile using the compilers installed on AutoLab. **Submissions that do not compile will be given a zero grade.**

The course assignments and project are to be completed by yourself. You may, however, discuss high level ideas on the course Piazza page. **Source code may not be shared or posted.**

Late policy for homework submissions

Late homework submissions are not accepted. We will use AutoLab for grading and you may submit as many times as you want prior to the due date.

2. Exam component (40% of final course grade - midterm 15% final 25%)

There will be one examination given during the semester and one final examination at the end of the term. You must attend all examinations.

The one in-semester exam will be held in lecture, (tentatively) during Week 7 of class, unless otherwise announced by the instructor in class.

The **final examination** will be given during the final exam period. The university schedules final examinations. Please check the HUB for the date, time and place of the final exam. If you have a conflict please let us know so we can schedule an alternative time for you to take the exam.

See <http://blogs.advising.buffalo.edu/beadvised/have-you-checked-your-final-exam-schedule>

If you miss an examination because of sickness or similar reasons, written notice and acceptable written documentation must be provided, as specified in the University Catalog. In the case of illness, you **MUST** visit a physician and obtain a note detailing the period during which you were medically incapable of taking the exam. Notify us as early as possible in writing (email is acceptable) if you miss an exam, before the exam takes place unless medically impossible. See your instructor as soon as you return to class.

If you miss an examination without a valid excuse, you will receive a zero grade for that examination. No make-up examinations will be available without a valid excuse.

You must bring a valid form of picture ID with you to each examination (a UB Card is preferred).

3. Active Learning component (10% of final course grade)

You will receive 1 participation point for each Top Hat question answered as well as 1 participation point for each recitation attended. TopHat will account for 50% of the participation grade and recitation attendance for 50% of the participation.

Overall course grade

The following table indicates the number to letter grade mapping we will use to assign letter grades at the end of the course. We reserve the right to make adjustments to the letter grade cut-offs. If any adjustments are made cut-offs will only be adjusted lower (e.g. the cut-off for an A may be moved from 90 to 89), never higher. The Grade points column is included for your convenience only, and is not official information. The official mapping can be found in the Undergraduate Catalog. Note that the minimum passing grade, for the course as a whole is 59% (unless adjusted).

Percentage score	Letter grade	Grade points
91-100	A	4.0
88-90	A-	3.67
83-87	B+	3.33
78-82	B	3.0
73-77	B-	2.67
68-72	C+	2.33
63-67	C	2.0
59-62	C-	1.67
0-58	F	0.0

Regrading

If you have a question about the grading of any piece of work, first consult with the teaching assistant who graded your work. If you cannot resolve your questions with the teaching assistant, you should consult with the instructor of the course.

Any questions about the grading of a piece of work ***must be raised within one week of the date that the work was returned*** by the teaching assistant or the instructor.

In other words, if you do not pick up your work in a timely fashion, you may forfeit your right to question the grading of your work.

Incomplete (I) grades

We will follow the UB Undergraduate Catalog Statement on Incomplete Grades, found in the Undergraduate Catalog.

Generally, incomplete ("I") grades are not given. However, very rarely, circumstances truly beyond a student's control prevents him or her from completing work in the course. In such cases the instructor can give a grade of "I". The student will be given instructions and a deadline for completing the work, usually no more than 30 days past the end of the semester. University and department policy dictate that "I" grades can be given only if the following conditions are met:

- An Incomplete will only be given for missing a small part of the course.
- An Incomplete will only be given when the student misses work due to circumstances beyond their control.
- An Incomplete will only be given when the student is passing the course except for the missed material.
- An Incomplete is to be made up with the original course instructor within the time specified by the appropriate University regulation (see appropriate document above), and usually within the following semester.
- An Incomplete will not be given to allow the student to informally retake the entire course, and have that grade count as the grade of the original course.

Incompletes cannot be given as a shelter from poor grades. **It is your responsibility to make a timely resignation from the course if you are doing poorly for any reason.**

General Notes

If you don't understand something covered in class, ask about it right away. The only silly question is the one not asked. If you get a poor mark on an assignment, quiz, or exam, find out why right away. Don't wait a month before asking. The course staff are available to answer your questions. Don't be afraid to approach the instructor or SA in class, during office hours, on Piazza, or through email.

Critical Campus Resources

Accessibility Resources

If you have any disability which requires reasonable accommodations to enable you to participate in this course, please contact the Office of Accessibility Resources in 60 Capen Hall, 716-645-2608 and also the instructor of this course during the first week of class. The office will provide you with information and review appropriate arrangements for reasonable accommodations, which can be found on the web at: <http://www.buffalo.edu/studentlife/who-we-are/departments/accessibility.html>.

Sexual Violence

UB is committed to providing a safe learning environment free of all forms of discrimination and sexual harassment, including sexual assault, domestic and dating violence and stalking. If you have experienced gender-based violence (intimate partner violence, attempted or completed sexual assault, harassment, coercion, stalking, etc.), UB has resources to help. This includes academic accommodations, health and counseling services, housing accommodations, helping with legal protective orders, and assistance with reporting the incident to police or other UB officials if you so choose. Please contact UB's Title IX Coordinator at 716-645-2266 for more information. For confidential assistance, you may also contact a Crisis Services Campus Advocate at 716-796-4399.

Mental Health

As a student you may experience a range of issues that can cause barriers to learning or reduce your ability to participate in daily activities. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, health concerns, or unwanted sexual experiences. Counseling, Health Services, and Health Promotion are here to help with these or other issues you may experience. You can learn more about these programs and services by contacting:

Counseling Services:

- 120 Richmond Quad (North Campus), 716-645-2720
- 202 Michael Hall (South Campus), 716-829-5800

Health Services:

- 4350 Maple Rd, Amherst, NY 14226, 716-829-3316

Health Promotion:

- 114 Student Union (North Campus), 716-645-2837

Diversity

The UB School of Engineering and Applied Sciences considers the diversity of its students, faculty, and staff to be a strength, critical to our success. We are committed to providing a safe space and a culture of mutual respect and inclusiveness for all. We believe a community of faculty, students, and staff who bring diverse life experiences and perspectives leads to a superior working environment, and we welcome differences in race, ethnicity, gender, age, religion, language, intellectual and physical ability, sexual orientation, gender identity, socioeconomic status, and veteran status.

Academic Integrity and Usage of AI

The full policy can be found here:

<https://engineering.buffalo.edu/computer-science-engineering/information-for-students/policies/academic-integrity.html>

The academic degrees and the research findings produced by our Department are worth no more than the integrity of the process by which they are gained. If we do not maintain reliably high standards of ethics and integrity in our work and our relationships, we have nothing of value to offer one another or to offer the larger community outside this Department, whether potential employers or fellow scholars.

For this reason, the principles of Academic Integrity have priority over every other consideration in every aspect of our departmental life, and we will defend these principles vigorously. It is essential that every student be fully aware of these principles, what the procedures are by which possible violations are investigated and adjudicated, and what the punishments for these violations are. Wherever they are suspected, potential violations will be investigated and determinations of fact sought. In short, breaches of Academic Integrity will not be tolerated.

Departmental Statement on Academic Integrity in Coding Assignments and Projects

The following statement further describes the specific application of these general principles to a common context in the CSE Department environment, the production of source code for project and homework assignments. It should be thoroughly understood before undertaking any cooperative activities or using any other sources in such contexts.

All academic work must be your own. Plagiarism, defined as copying or receiving materials from a source or sources and submitting this material as one's own without acknowledging the particular debts to the source (quotations, paraphrases, basic ideas), or otherwise representing the work of another as one's own, is never allowed. Collaboration, usually evidenced by unjustifiable similarity, is never permitted in individual assignments. Any submitted academic work may be subject to screening by software programs designed to detect evidence of plagiarism or collaboration.

It is your responsibility to maintain the security of your computer accounts and your written work. Do not share passwords with anyone, nor write your password down where it may be seen by others. Do not change permissions to allow others to read your course directories and files. Do not walk away from a workstation without logging out. These are your responsibilities. In groups that collaborate inappropriately, it may be impossible to determine who has offered work to others in the group, who has received work, and who may have inadvertently made their work available to the others by failure to maintain adequate personal security. In such cases, all will be held equally liable.

These policies and interpretations may be augmented by individual instructors for their courses. Always check the handouts and web pages of your course and section for additional guidelines.

Departmental and Course Policy on Violations of Academic Integrity

The CSE Department has a zero-tolerance policy regarding academic integrity (AI) violations.

When there is a potential violation of academic integrity in a course, the course director shall first notify the concerned students.

This notification begins the review and appeals process defined in the University's Academic Integrity statement:

<http://catalog.buffalo.edu/policies/course/integrity.html>

Upon conclusion of the review and appeals process, if the department, school, and university have determined that the student has committed a violation, the following sanctions will be imposed upon the student:

§ 1. **Documentation.** The department, school, and university will record the student's name in departmental, decanal, and university-level academic integrity violations databases.

§ 2. **Penalty Assessment.** The standing policy of the Department is that all students involved in an academic integrity violation will receive an F grade for the course. The course director may recommend a lesser penalty for the first instance of academic integrity violation, and the adjudication committees that hear the appeal at the department, decanal and provost level may recommend a lesser or greater penalty.

Usage of AI

Usage of AI to generate code is not allowed in the course and will result in an F. You can use AI assistive technology like co-pilot for robust line completion only. You are, however, responsible for understanding any lines of code in your program. To determine incorrect usage of AI in the course, you may be asked to explain your code in detail. Failure to be able to do so will result in an academic integrity violation.

Amnesty Policy

We understand that students are under a lot of pressure and people make mistakes. If you have concerns that you may have violated academic integrity on a particular assignment, and would like to withdraw the assignment, you may do so by sending us an email **BEFORE THE VIOLATION IS DISCOVERED BY US**. The email should take the following format:

Dear Dr. Mikida and Dr. Ye,

I wish to inform you that on assignment X, the work I submitted was not entirely my own. I would like to withdraw my submission from consideration to preserve academic integrity.

J.Q. Student
Person #12345678
UBIT: jqstuden

When we receive this email, student J would receive a 0 on assignment X, but would not receive an F for the course, and would not be reported to the office of academic integrity.